

# Architecture Validation & Quality Report

## 1. Executive Summary

Overall Architecture Validation Result: **SUCCESS**

## 2. Structural Analysis

| Metric                     | Observed Value |
|----------------------------|----------------|
| Component Count            | 4              |
| Relationship Count         | 3              |
| Undefined References       | 0              |
| Self Loops                 | 0              |
| Invalid Relationship Types | 0              |

### Metric Definitions and Calculation Rules

Component Count: The total number of components explicitly declared in the architectural description.

Relationship Count: The total number of relationships declared between components.

Undefined References: The number of relationships whose source or target does not match any declared component name.

Self-Loops: The number of relationships where the source and target components are identical.

Invalid Relationship Types: The number of relationships whose interaction type is not recognized by the validation rules.

### Structural Evaluation Interpretation

Structural validation assesses the syntactic and referential correctness of the architecture. Any non-zero value in Undefined References, Self-Loops, or Invalid Relationship Types indicates an invalid architectural description.

## 3. Quality Analysis

| Component      | Fan-In | Fan-Out | Coupling |
|----------------|--------|---------|----------|
| User Service   | 1      | 0       | 1        |
| Order Database | 1      | 0       | 1        |
| API Gateway    | 0      | 2       | 2        |
| Order Service  | 1      | 1       | 2        |

### Calculation Formulae

Fan-In(component) = Number of relationships where component is the target.

Fan-Out(component) = Number of relationships where component is the source.

Coupling(component) = Fan-In(component) + Fan-Out(component).

### Quality Evaluation Interpretation

Components with high coupling are strongly dependent on other components. High coupling increases maintenance effort, change propagation, and the risk of cascading failures across the architecture.

## 4. Domain Analysis

| Metric              | Observed Value |
|---------------------|----------------|
| Stateless Ratio     | 1.0            |
| State Concentration | 1              |
| Communication Style | synchronous    |

### Calculation Rules

Stateless Ratio: Inferred value representing the proportion of components without explicitly declared state.

State Concentration: Maximum Fan-In value observed across all components.

Communication Style: Determined by comparing the number of event-driven relationships to synchronous data-flow relationships.

### Domain Evaluation Interpretation

High state concentration and synchronous communication patterns indicate centralized coordination and strong runtime dependencies, which negatively impact scalability, resilience, and fault tolerance.

## 5. Suggested Improvements

- Synchronous Communication Pattern: Components are dependent on immediate responses from other components at runtime.